

Ayman Al Rubaey

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Education

University of Victoria, Victoria, BC
Bachelor of Engineering, **Computer Engineering**

Sep 2021 - May 2026

Experience

Electronics/Mechatronics Engineer (Co-op and Contractor)

May 2025 - Present

Telix ARTMS Inc, Vancouver, BC

- Designed and debugged custom PCB schematics in KiCad, performing precision soldering of SMT and through-hole components for functional prototypes.
- Designed electrical schematics and wiring diagrams using SolidWorks Electrical for integration with mechanical systems. Created and maintained detailed assembly documentation, such as wiring diagrams, schematics, BOMs, and step-by-step instructions to standardize processes.
- Programmed and configured PLCs using TIA Portal and designed HMI interface using WinCC to enable real-time system monitoring and control.

Software Product Designer (Co-op)

Sep 2023 - Dec 2023

Dometic, Vancouver, BC

- Developed and implemented C/C++ and Python functions, along with test scripts, to rigorously verify software functionality and ensure product reliability.
- Created a Python program to control a power supply by sending CAN messages to automate testing for the quality assurance team.
- Implemented a vibration sensor driver on STM32 for an accelerometer using SPI communication, enabling vibration monitoring and early fault detection.

Embedded Software QA Engineer (Co-op)

Jan 2023 - Apr 2023

Lantronix, Vancouver, BC

- Conducted software validation for an electric car project called TOGG, including infotainment and connectivity systems, ensuring reliable in-vehicle user experience.
- Utilized CANoe, CANalyzer, and a logic analyzer to validate communication protocols and troubleshoot embedded system issues.
- Developed Python scripts to automate testing and documented test cases to support reproducible and efficient QA processes.

Electrical Assembly Technician (Co-op)

May 2022 - Aug 2022

Charge Solar, Victoria, BC

- Assembled and wired custom electric power panels by interpreting electrical schematics and ensuring compliance with customer specifications.
- Fabricated a variety of cable types following detailed work orders and supervisor instructions.
- Performed continuity and insulation testing on assembled panels to verify wiring integrity and safety.

Projects

16 Bit CPU in VHDL

- Designed and implemented a custom 16-bit RISC processor using VHDL, featuring a five-stage pipelined architecture (Fetch, Decode, Execute, Memory, Writeback). The CPU supports a wide range of instructions including arithmetic, logical, memory access, and branch operations.
- Integrated hazard detection logic to handle data hazards within the pipeline. Verified instruction execution and pipeline behavior through simulation and testbenches, ensuring correct operation of different instructions.

Traffic Light System

- Built the physical circuit for the traffic light system, wiring LEDs, potentiometer, and shift registers to the STM32F4 board.
- Programmed a real-time Traffic Light System in C for STM32F4 using FreeRTOS, implementing multitasking with queues and timers, ADC based traffic flow control, and LED vehicle simulation through shift register manipulation.

PWM Signal Generation and Monitoring System

- Programmed an embedded system in C on a STM32F0 microcontroller to monitor and control a PWM signal from a 555 timer, adjusting frequency through an optocoupler using ADC and DAC.
- Implemented SPI protocol to display real-time frequency and potentiometer resistance on an OLED display, with interrupt-driven toggling between 555 timer and Function Generator signal measurements.

Digital Alarm Clock with Radio

- Developed the schematic and PCB layout for the project's electronic components using KiCad, and designed a custom PCB enclosure in SolidWorks considering both form and function
- Implemented a Python application on a Raspberry Pi Pico to interface with hardware components, using SPI for the OLED display and I2C for the radio chip, enabling radio tuning, volume control, and real-time information display

Autonomous IR Detection Robot

- Designed a fully autonomous robot that navigates in an arena to locate a target that emits IR waves and drops an object on top by using components in a VEX kit.
- Utilized expertise in both electrical and software engineering to contribute to the development of the robot, and programmed its autonomous movement using C language.

Skills

Software Skills: C/C++, Python, MATLAB, Assembly, VHDL (FPGA), TIA Portal, VScode, Git, Linux, QNX, SQL, JIRA, FreeRTOS, Wireshark

Electrical & Design Skills: Circuit Design (LTSpice), KiCad (PCB Design), Solidworks, Solidworks Electrical, KeysightADS, CANoe/CANalyzer, Soldering, Oscilloscope, Function Generator, DMM